

Aim

To prepare the Software Requirement Specification document for the approved project

Software Requirements Specification (SRS)

Project Title

TransitOps – Smart Transport Operations Platform

Version: 1.0

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1. Introduction

Transportation and logistics companies play a vital role in the movement of goods and services. Many organizations still depend on spreadsheets, paper records, and manual processes to manage vehicles, drivers, trip scheduling, maintenance, and operational expenses. These traditional methods often lead to scheduling conflicts, underutilized vehicles, delayed maintenance, expired driver licenses, inaccurate expense tracking, and poor decision-making due to a lack of real-time information.

TransitOps is a web-based transport operations platform designed to digitize and automate fleet management. The system centralizes vehicle registration, driver management, trip scheduling, maintenance tracking, fuel and expense management, and reporting into a single platform. It also enforces business rules to ensure safe, efficient, and reliable transport operations.

2. Purpose

The purpose of TransitOps is to provide a centralized transport management system that automates daily fleet operations while improving operational efficiency, data accuracy, and decision-making. The application aims to eliminate manual record-keeping and provide role-based access for different users involved in transport management.

3. Scope

The system provides an integrated platform for managing transport operations within logistics companies, courier services, fleet operators, and organizations that own multiple vehicles.

The system allows users to:

- Register and manage vehicles.
- Maintain driver records.
- Create and dispatch trips.
- Track maintenance activities.
- Record fuel usage and operational expenses.
- Monitor fleet performance through dashboards.
- Generate operational reports.
- Manage users through Role-Based Access Control (RBAC).

The project focuses on internal transport operations and does not include customer booking or online payment services.

4. Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
RBAC	Role-Based Access Control
ROI	Return on Investment
KPI	Key Performance Indicator
CRUD	Create, Read, Update and Delete
Fleet	Collection of vehicles owned by an organization
Dispatcher	Person responsible for assigning trips
Odometer	Vehicle distance measurement
JWT	JSON Web Token
MERN	MongoDB, Express.js, React.js, Node.js

5. Overall Description

TransitOps is a web application that enables organizations to manage transport operations digitally. Users access the system according to their assigned roles. The application maintains vehicle and driver information, schedules trips, tracks maintenance, records fuel expenses, and generates analytical reports.

The system automatically validates operational rules such as vehicle availability, driver eligibility, and cargo capacity before dispatching trips. Dashboard analytics provide managers with real-time operational insights.

6. User Characteristics

6.1 Administrator

Responsibilities

- Manage user accounts
 - Assign user roles
 - Configure system settings
 - Monitor system activities
-

6.2 Fleet Manager

Responsibilities

- Register vehicles
 - Manage maintenance schedules
 - Monitor fleet utilization
 - Track vehicle lifecycle
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6.3 Dispatcher

Responsibilities

- Create trips
 - Assign vehicles
 - Assign drivers
 - Monitor active trips
-

6.4 Safety Officer

Responsibilities

- Verify driver licenses
 - Monitor safety scores
 - Track license expiry
 - Ensure compliance
-

6.5 Financial Analyst

Responsibilities

- Monitor fuel expenses
 - Analyze maintenance costs
 - Calculate vehicle ROI
 - Generate financial reports
-

7. Functional Requirements

R-1 User Authentication

Description

The system shall authenticate users and provide secure access based on their assigned roles.

R-1.1 User Login

Description

The system shall allow registered users to log in using their email address and password.

Input

- Email Address
- Password

Processing

- Validate user credentials.
- Verify password.

- Check whether the user account is active.

Output

- Successful login redirects the user to the dashboard.
- Invalid credentials display an error message.

R-1.2 Role-Based Access Control

Description

The system shall grant access to modules based on the user's assigned role.

Input

- User Role

Processing

- Identify the logged-in user's role.
- Load only the permitted modules.

Output

- Authorized modules are displayed.
- Unauthorized modules remain inaccessible.

R-1.3 Logout

Description

The system shall allow authenticated users to log out securely.

Input

- Logout Request

Processing

- Destroy user session.
- Remove authentication token.

Output

- User is redirected to the login page.
-

R-2 Dashboard

Description

The dashboard provides real-time operational statistics.

R-2.1 Dashboard KPIs

Input

Vehicle, Driver and Trip Records

Processing

Calculate:

- Active Vehicles
- Available Vehicles
- Vehicles in Maintenance
- Active Trips
- Pending Trips
- Drivers on Duty
- Fleet Utilization

Output

Dashboard displays updated KPIs.

R-2.2 Dashboard Filters

Input

- Vehicle Type
- Status
- Region

Processing

Filter dashboard data.

Output

Filtered dashboard statistics.

R-2.3 Recent Activities

Input

System activity logs.

Processing

Retrieve latest activities.

Output

Display recent operations performed.

R-3 Vehicle Management

Description

Manage all fleet vehicles.

R-3.1 Add Vehicle

Input

- Registration Number
- Vehicle Name
- Vehicle Type
- Maximum Load Capacity
- Odometer
- Acquisition Cost

Processing

- Validate input.
- Check registration uniqueness.
- Save vehicle.

Output

Vehicle successfully registered.

R-3.2 Update Vehicle

Input

Updated vehicle details.

Processing

Modify vehicle information.

Output

Vehicle information updated.

R-3.3 Delete Vehicle

Input

Vehicle ID

Processing

Delete retired vehicle.

Output

Vehicle removed successfully.

R-3.4 View Vehicle

Input

Vehicle ID

Processing

Retrieve vehicle details.

Output

Complete vehicle information displayed.

R-3.5 Update Vehicle Status

Input

Vehicle Status

Processing

Update status according to trip or maintenance.

Output

Current status displayed.

R-4 Driver Management

Description

Manage driver records and eligibility.

R-4.1 Register Driver

Input

- Name
- License Number
- License Category
- Contact Number
- License Expiry Date

Processing

Validate and store driver details.

Output

Driver successfully registered.

R-4.2 Update Driver

Input

Updated driver information.

Processing

Modify driver record.

Output

Driver details updated.

R-4.3 Suspend Driver

Input

Driver ID

Processing

Change status to Suspended.

Output

Driver becomes unavailable.

R-4.4 View Driver

Input

Driver ID

Processing

Retrieve driver information.

Output

Driver profile displayed.

R-5 Trip Management

Description

Manage transport trips.

R-5.1 Create Trip

Input

- Source
- Destination
- Vehicle
- Driver
- Cargo Weight
- Planned Distance

Processing

Validate all business rules.

Output

Trip created.

R-5.2 Assign Vehicle

Input

Vehicle ID

Processing

Check vehicle availability.

Output

Vehicle assigned.

R-5.3 Assign Driver

Input

Driver ID

Processing

Verify driver's license and availability.

Output

Driver assigned.

R-5.4 Dispatch Trip

Input

Dispatch request.

Processing

Update trip status to **Dispatched**.

Change vehicle and driver status to **On Trip**.

Output

Trip dispatched successfully.

R-5.5 Complete Trip

Input

Trip completion details.

Processing

Update trip.

Change vehicle and driver status to **Available**.

Output

Trip completed.

R-5.6 Cancel Trip

Input

Trip ID

Processing

Cancel trip.

Restore driver and vehicle availability.

Output

Trip cancelled.

R-6 Maintenance Management

Description

Maintain vehicle servicing records.

R-6.1 Create Maintenance

Input

Vehicle ID

Maintenance Type

Processing

Create maintenance record.

Update vehicle status to **In Shop**.

Output

Maintenance record created.

R-6.2 Close Maintenance

Input

Maintenance ID

Processing

Close maintenance.

Restore vehicle availability.

Output

Maintenance completed.

R-6.3 View Maintenance History

Input

Vehicle ID

Processing

Retrieve maintenance records.

Output

Maintenance history displayed.

R-7 Fuel & Expense Management

Description

Manage operational expenses.

R-7.1 Fuel Log

Input

Vehicle

Fuel Quantity

Fuel Cost

Date

Processing

Store fuel record.

Output

Fuel log created.

R-7.2 Expense Log

Input

Expense Type

Amount

Date

Processing

Store expense.

Output

Expense recorded.

R-7.3 Calculate Operational Cost

Input

Fuel and Maintenance records.

Processing

Calculate total operational cost.

Output

Operational cost displayed.

R-8 Reports & Analytics

Description

Generate transport analytics.

R-8.1 Fuel Efficiency Report

Input

Trip and Fuel records.

Processing

Calculate fuel efficiency.

Output

Fuel efficiency report.

R-8.2 Fleet Utilization Report**Input**

Vehicle records.

Processing

Calculate utilization percentage.

Output

Fleet utilization report.

R-8.3 Vehicle ROI Report**Input**

Revenue, Fuel Cost, Maintenance Cost.

Processing

Calculate ROI.

Output

Vehicle ROI displayed.

R-8.4 Export Reports**Input**

Export request.

Processing

Generate CSV (and optionally PDF).

Output

Report downloaded.

R-9 Business Rules

Description

The system shall enforce transport operation rules.

R-9.1 Vehicle Validation

Input

Vehicle selection.

Processing

Verify vehicle is available and not under maintenance.

Output

Eligible vehicle selected.

R-9.2 Driver Validation

Input

Driver selection.

Processing

Check license validity and status.

Output

Eligible driver selected.

R-9.3 Cargo Validation

Input

Cargo Weight

Vehicle Capacity

Processing

Compare cargo weight with maximum capacity.

Output

Trip approved or rejected.

R-9.4 Automatic Status Updates

Input

Trip or Maintenance Status

Processing

Automatically update Vehicle and Driver status.

Output

Current status reflected throughout the system.

8. Non-Functional Requirements

Performance

- System response time should be less than 3 seconds.
- Dashboard data should load quickly.

Security

- Passwords shall be encrypted.
- Authentication shall use JWT.
- RBAC shall restrict unauthorized access.

Reliability

- Data shall remain consistent.
- System shall prevent duplicate vehicle registration.

Availability

- System should be available 24×7 except scheduled maintenance.

Scalability

- The system shall support increasing numbers of vehicles and users.

Usability

- User interface shall be simple and responsive.
- Navigation shall be easy to understand.

Maintainability

- Code shall follow modular architecture.
 - Future modules can be added easily.
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9. Constraints

- Internet connection is required.
 - Users must have valid login credentials.
 - Only authorized users can access restricted modules.
 - The project will be developed using the MERN Stack.
 - MongoDB will be used as the primary database.
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10. Assumptions

- Every vehicle has a unique registration number.
 - Every driver possesses a valid driving license.
 - Users have basic computer knowledge.
 - Internet connectivity is available.
 - Organization policies are followed while using the system.
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11. Software Requirements

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT (JSON Web Token)

Development Tools

- Visual Studio Code
- Postman

Version Control

- Git
 - GitHub
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12. Hardware Requirements

Server

- Processor: Intel Core i5 or above
- RAM: 8 GB minimum
- Storage: 100 GB or above

Client

- Processor: Intel Core i3 or above
 - RAM: 4 GB minimum
 - Browser: Google Chrome, Microsoft Edge, Mozilla Firefox
 - Internet Connection
-

13. User Interface Requirements

The application shall provide:

- Login Page
- Dashboard
- Vehicle Management Page
- Driver Management Page
- Trip Management Page
- Maintenance Management Page
- Fuel & Expense Page
- Reports Dashboard
- User Profile
- Responsive Design for Desktop and Mobile

The interface shall be simple, user-friendly, and easy to navigate.

14. Future Enhancements

The following features can be added in future versions:

- GPS-based live vehicle tracking.
- Route optimization using AI.
- Predictive maintenance using machine learning.
- Mobile application for Android and iOS.
- SMS and Email notifications.
- QR code-based vehicle verification.
- Integration with IoT sensors.
- Real-time traffic monitoring.
- Cloud backup and disaster recovery.
- Advanced analytics and business intelligence dashboards.